WASP-12b

R-0545

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-12_181112` · created 2026-07-22T04:58:32+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2458434.8126 +/- 0.00055 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1396 +/- 0.0083
Transit depth (Rp/Rs)^2	1.95 +/- 0.23 [%]
Orbital Inclination (inc)	79.91 +/- 1.44
Airmass coefficient 1 (a1)	1.05 +/- 0.011
Airmass coefficient 2 (a2)	-0.0452 +/- 0.0042
Scatter in the residuals of the lightcurve fit is	1.02 %
Best Comparison Star	#2 - [241, 142]
Optimal Aperture	4.38
Optimal Annulus	10.22
Transit Duration (day)	0.119 +/- 0.0044

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1396 ± 0.0083 vs 0.1178 (deviation 2.32σ)
Duration fit vs published	0.119 d vs 0.125 d (deviation 1.21σ)
Mid-time O–C	-1.3 min
Depth SNR	14.9
Residual scatter	1.02 %

Light curve

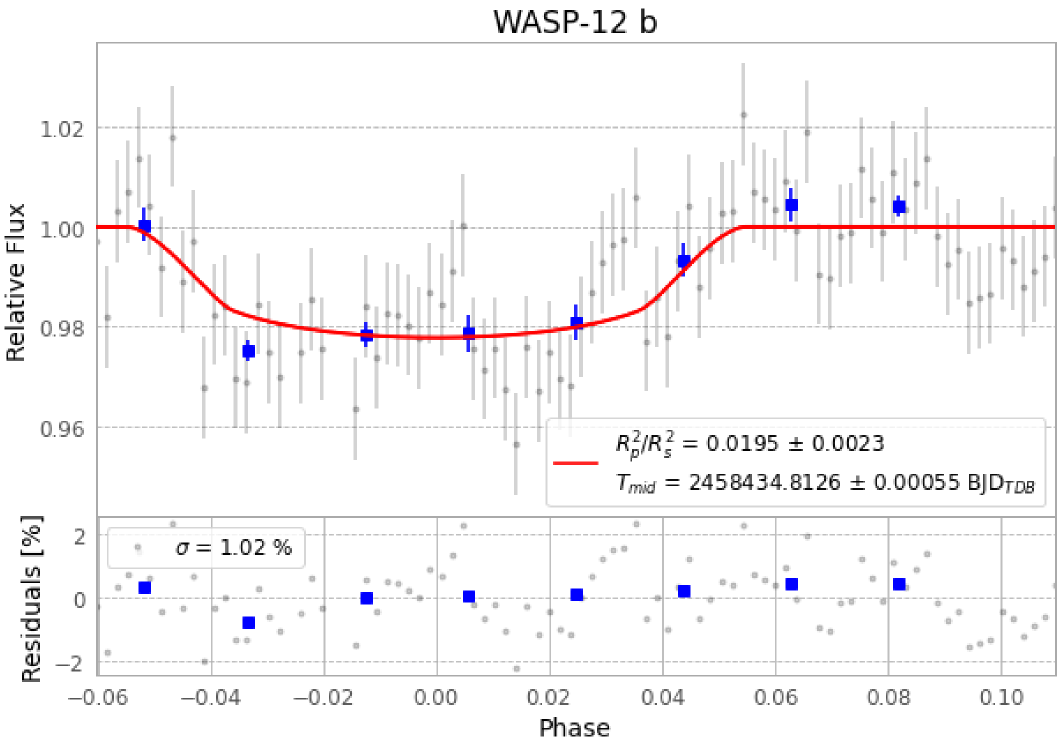

Figure 1 — FinalLightCurve_WASP-12 b_2018-11-11.png (SHA-256 4af742fb9d678ea4…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [353, 159], 2 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 adcd9b4420febb6d…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494`

Data provenance

88 FITS frames (58 MB), `WASP-12181112054812.FITS` → `WASP-12181112101512.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-181112.log (78fd0cc40bf6…), FinalLightCurve_WASP-12 b_2018-11-11.png (4af742fb9d67…), FinalLightCurve_WASP-12 b_2018-11-11.csv (9be4e9abaec0…)

Compute resources

Wall time 105.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)